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Fluid connection assembly

Abstract

A fluid connection assembly, including a connector body, including a first end, a second end, a through-bore, a first radially outward facing surface including an annular groove, a first radially inward facing surface extending from the first end, and a second radially inward facing surface extending from the second end, and a retainer operatively arranged to be removably connected to the connector body, the retainer including a third end, a fourth end, a first section, a second section displaceable with respect to the first section, a third radially inward facing surface, a first flange extending radially inward from the third radially inward facing surface and operatively arranged to engage the annular groove, and a second flange extending radially inward from the third radially inward facing surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is the U.S. National Stage Application pursuant to 35 U.S.C. § 371 of International Patent Application No. PCT/US2022/071545, filed on Apr. 5, 2022, which application claims the benefit under Articles 4 and 8 of the Stockholm Act of the Paris Convention for the Protection of Industrial Property of International Patent Application No. PCT/US2021/071953, filed on Oct. 21, 2021, and International Patent Application No. PCT/US2021/035393, filed on Jun. 2, 2021, which applications are hereby incorporated by reference herein in their entireties.

FIELD

(2) The present disclosure relates to fluid connectors, and more particularly, to a fluid connection assembly for a refrigeration system including a connector body and a retainer that decreases the insertion force required for assembly and allows for quick assembly of components without the need for tools.

BACKGROUND

(3) Fluid connectors, fluid connections, and fluid connection assemblies are integral components for many applications, and especially for automotive and industrial applications. Since refrigeration systems are made up of various components such as a compressor, air cooler, condenser, receiver, evaporator or chiller, and/or suction drum, fluid must be able to travel not only within each

component but also between components. Fluid predominantly moves between components via flexible or rigid hoses which connect to each component by fluid connectors. Such fluid connectors typically include a retaining clip, retaining ring clip, or snap ring carried on the connector body which is adapted to snap behind a raised shoulder of a tube when the tube is fully inserted into the connector body. However, in order for the fluid connector to properly function, slots or apertures must be machined in the connector body such that the retaining clip can protrude therethrough and engage the tube, which requires extra post-process manufacturing. During the assembly process, installation of the retaining clip onto the connector body is difficult and failure to install the retaining clip properly can jeopardize the structural integrity of the retaining clip. Additionally, the force required to engage the tube into the connector body, and overcome the radial force of the retaining clip, is very large with current designs. Also, since the retaining clips are very thin and small, it is easy to lose them if dropped or misplaced. Furthermore, some connection assembly solutions take a long time to secure and require tools for the assembly process. Moreover, current connection assembly solutions do not indicate the connection state of the assembly.

(4) Thus, there has been a long-felt need for a fluid connection assembly for a refrigeration system including a connector body and a retainer that allows for quick assembly, disassembly, eliminates the need for post-process machining, and reduces the insertion force required to assemble the fluid connector.

SUMMARY

(5) According to aspects illustrated herein, there is provided a fluid connection assembly, comprising a connector body, comprising a first end, a second end, a through-bore, a first radially outward facing surface comprising an annular groove, a first radially inward facing surface extending from the first end, and a second radially inward facing surface extending from the second end, and a retainer operatively arranged to be removably connected to the connector body, the retainer including a third end, a fourth end, a first section, a second section displaceable with respect to the first section, a third radially inward facing surface, a first flange extending radially inward from the third radially inward facing surface and operatively arranged to engage the annular groove, and a second flange extending radially inward from the third radially inward facing surface.

(6) In some embodiments, the first radially inward facing surface comprises at least one annular groove. In some embodiments, the at least one annular groove comprises a first annular groove, and a second annular groove spaced apart in an axial direction from the first annular groove. In some embodiments, a seal is arranged in the at least one annular groove. In some embodiments, the second radially inward facing surface comprises threading. In some embodiments, the connector body further comprises a fourth radially inward facing surface arranged between first radially inward facing surface and the second radially inward facing surface, wherein the diameter of the fourth radially inward facing surface is less than both the first radially inward facing surface and the second radially inward facing surface. In some embodiments, the connector body further comprises an axial surface connecting the first radially inward facing surface and the fourth radially inward facing surface. In some embodiments, the connector body further comprises a frusto-conical surface connecting the second radially inward facing surface and the fourth radially inward facing surface.

(7) In some embodiments, the first flange is arranged immediately proximate the third end, and the second flange is arranged immediately proximate the fourth end. In some embodiments, the first flange comprises a first diameter, the second flange comprises a second diameter, and the first diameter is greater than the second diameter. In some embodiments, the fluid connection assembly further comprises a tube including a shoulder, wherein the retainer is operatively arranged to secure the tube to the connector body. In some embodiments, the fluid connection assembly further comprises a service valve body comprising a plurality of ports, wherein the connector body is operatively arranged to be removably connected to at least one port of the plurality of ports.

(8) According to aspects illustrated herein, there is provided a fluid connection assembly, comprising a connector body, comprising a first end, a second end, a through-bore extending from the first end to the second end, a first radially outward facing surface comprising an annular groove, a first radially inward facing surface extending from the first end, a second radially inward facing surface extending from the second end, and a third radially inward facing surface arranged between the first radially inward facing surface and the second radially inward facing surface, the third radially inward facing surface having a diameter that is less than the diameter of both the first radially inward facing surface and the second radially inward facing surface.

(9) In some embodiments, the fluid connection assembly further comprises a retainer operatively arranged to be removably connected to the connector body, the retainer including a third end, a fourth end, a first section, a second section displaceable with respect to the first section, a fourth radially inward facing surface, a first flange extending radially inward from the fourth radially inward facing surface and operatively arranged to engage the annular groove, and a second flange extending radially inward from the fourth radially inward facing surface. In some embodiments, the first flange is arranged immediately proximate the third end, and the second flange is arranged immediately proximate the fourth end. In some embodiments, the first flange comprises a first diameter, the second flange comprises a second diameter, and the first diameter is greater than the second diameter. In some embodiments, the first radially inward facing surface comprises a first annular groove, and a second annular groove spaced apart in an axial direction from the first annular groove. In some embodiments, the second radially inward facing surface comprises threading. In some embodiments, the connector body further comprises an axial surface connecting the first radially inward facing surface and the third radially inward facing surface. In some embodiments, the connector body further comprises a frusto-conical surface connecting the second radially inward facing surface and the third radially inward facing surface.

(10) According to aspects illustrated herein, there is provided a service port quick connect body that improves the ease of connection for industrial refrigerant applications, as well as eliminates the need for braze or solder joints. The present disclosure reduces the potential for operator induced quality failures and/or leaks, and the amount of time to connect a system.

(11) The service port of the present disclosure comprises a quick connector body removably connectable thereto. For example, the quick connector body may be threadably engaged with the service port. The quick connector body comprises one or more O-rings and a flange to receive a plastic retainer. The tube bead abuts to the quick connect body flange and the retainer closes in on the bead and the quick connect body flange to retain the tube and prevent it from being removed. To service the system in the field, the retainer can be a replacement part provided by a service technician. Using a small pick/tool, the latch on the retainer is pried open.

(12) These and other objects, features, and advantages of the present disclosure will become readily apparent upon a review of the following detailed description of the disclosure, in view of the drawings and appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various embodiments are disclosed, by way of example only, with reference to the accompanying schematic drawings in which corresponding reference symbols indicate corresponding parts, in which:
- (2) FIG. 1A is a front perspective view of a fluid connection assembly;
- (3) FIG. 1B is a rear perspective view of the fluid connection assembly shown in FIG. 1A;
- (4) FIG. 2 is an exploded perspective view of the fluid connection assembly shown in FIG. 1A;
- (5) FIG. 3 is a perspective view of the retainer shown in FIG. 1A, in an unlocked state; and,

(6) FIG. 4 is a cross-sectional view of the fluid connection assembly taken generally along line 4-4 in FIG. 1A.

DETAILED DESCRIPTION

(7) At the outset, it should be appreciated that like drawing numbers on different drawing views identify identical, or functionally similar, structural elements. It is to be understood that the claims are not limited to the disclosed aspects.

(8) Furthermore, it is understood that this disclosure is not limited to the particular methodology, materials and modifications described and as such may, of course, vary. It is also understood that the terminology used herein is for the purpose of describing particular aspects only, and is not intended to limit the scope of the claims.

(9) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure pertains. It should be understood that any methods, devices or materials similar or equivalent to those described herein can be used in the practice or testing of the example embodiments. The assembly of the present disclosure could be driven by hydraulics, electronics, pneumatics, and/or springs.

(10) It should be appreciated that the term “substantially” is synonymous with terms such as “nearly,” “very nearly,” “about,” “approximately,” “around,” “bordering on,” “close to,” “essentially,” “in the neighborhood of,” “in the vicinity of,” etc., and such terms may be used interchangeably as appearing in the specification and claims. It should be appreciated that the term “proximate” is synonymous with terms such as “nearby,” “close,” “adjacent,” “neighboring,” “immediate,” “adjoining,” etc., and such terms may be used interchangeably as appearing in the specification and claims. The term “approximately” is intended to mean values within ten percent of the specified value.

(11) It should be understood that use of “or” in the present application is with respect to a “non-exclusive” arrangement, unless stated otherwise. For example, when saying that “item x is A or B,” it is understood that this can mean one of the following: (1) item x is only one or the other of A and B; (2) item x is both A and B. Alternately stated, the word “or” is not used to define an “exclusive or” arrangement. For example, an “exclusive or” arrangement for the statement “item x is A or B” would require that x can be only one of A and B. Furthermore, as used herein, “and/or” is intended to mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur. For example, a device comprising a first element, a second element and/or a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element.

(12) Moreover, as used herein, the phrases “comprises at least one of” and “comprising at least one of” in combination with a system or element is intended to mean that the system or element includes one or more of the elements listed after the phrase. For example, a device comprising at least one of: a first element; a second element; and, a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element. A similar interpretation is intended when the phrase “used in at least one of:” is used herein. Furthermore, as used herein, “and/or” is intended to mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur.

(13) It should be appreciated that the term “tube” as used herein is synonymous with hose, pipe, channel, conduit, tube end form, or any other suitable pipe flow used in hydraulics and fluid

mechanics. It should further be appreciated that the term “tube” can mean a rigid or flexible conduit of any material suitable for containing and allowing the flow of a gas or a liquid.

(14) “Fully assembled locked state” as used herein refers to the fluid connection assembly and indicates that the tube is fully connected to the service valve body and the retainer is fully connected to the service valve body to retain the tube therein. “Partially assembled locked state” as used herein refers to the fluid connection assembly and indicates that the retainer is fully connected to the service valve body but the tube is not arranged in the service valve body or the retainer. The retainer may also be described as having a “locked state” or an “unlocked state,” which indicates the state of the retainer independent of the service valve body and the tube.

(15) Adverting now to the figures, FIG. 1A is a front perspective view of fluid connection assembly 10. FIG. 1B is a rear perspective view of fluid connection assembly 10. FIG. 2 is an exploded perspective view of fluid connection assembly 10. Fluid connection assembly 10 generally comprises retainer 20, tube 80, connector body 40, and service valve body 100. The following description should be read in view of FIGS. 1A-2.

(16) Tube 80 comprises end 82, section 83, bead or shoulder 87, section 89, end 92, and through-bore 94. Through-bore 94 extends through tube 80 from end 82 to end 92. Section 183 is arranged between end 82 and shoulder 87 and comprises radially outward facing surface 84. Radially outward facing surface 84 includes a substantially constant diameter. In some embodiments, radially outward facing surface 84 comprises a frusto-conical taper or curvilinear surface proximate end 82 (see FIG. 4). Shoulder 87 is arranged between section 83 and section 89 and comprises surface 86 and surface 88. In some embodiments, surface 86 is an axial surface facing at least partially in axial direction AD1 and surface 88 is an axial surface facing at least partially in axial direction AD2. In some embodiments, surface 86 is a frusto-conical surface extending from the radially outward facing surface of shoulder 87 radially inward in axial direction AD1. For example, surface 86 may be a linear conical shape and increase in diameter in axial direction AD2. In some embodiments, surface 86 may comprise a linear portion and a conical or frusto-conical portion. Section 89 is arranged between shoulder 87 and end 92 and comprises radially outward facing surface 90. Radially outward facing surface 90 includes a substantially constant diameter. Tube 80 is arranged to be inserted, specifically with end 82 first, into connector body 40. Tube 80 is inserted into connector body 40 until section 83, or radially outward facing surface 84, engages seals 62A-B (see FIG. 4). Shoulder 87 engages end 44, more specifically, surface 86 abuts against and/or engages end 44, at which point retainer 20 is assembled to secure tube 80 to service valve body 100, as will be described in greater detail below. It should be appreciated that tube 80 may be any traditional tube or tube end form comprising a bead, radially outward extending protrusion or flange, or ramp profile, which extends radially outward and axially on the outer surface of the tube, to secure the tube within the connector body. In some embodiments, tube 80 comprises a metal, a polymer, and/or a ceramic.

(17) FIG. 3 is a perspective view of retainer 20, in an unlocked state. Retainer 20 is shown in FIGS. 1A-2 in a locked state. FIG. 4 is a cross-sectional view of fluid connection assembly 10 taken generally along line 4-4 in FIG. 1A. The following description should be read in view of FIGS. 1A-4.

(18) Retainer 20 generally comprises section 20A, section 20B, end 22, end 24, hole 21 extending from end 22 to end 24, radially outward facing surface 26, and radially inward facing surface 28. In some embodiments, section 20A is hingedly connected to section 20B, via, for example, hinge 34. In some embodiments, hinge 34 is a living hinge. In some embodiments, section 20A is removably connected to section 20B. When sections 20A and 20B are connected or retainer 20 is in the locked state (FIGS. 1A-2), hole 21 is formed therebetween; however, it should be appreciated that even in the unlocked state (FIG. 3), it could be said that each of sections 20A and 20B include a respective hole 21.

(19) Radially inward facing surface 28 extends from end 22 to end 24 and is operatively arranged

to engage radially outward facing surface **52**. Radially inward facing surface **28** comprises flange **30** extending radially inward therefrom. In some embodiments, flange **30** is arranged immediately adjacent end **22**. Flange **30** is operatively arranged to engage groove **54** and radially inward facing surface **28** is operatively arranged to engage radially outward facing surface **52** to connect retainer **20** to connector body **40**. Flange **30** comprises diameter D1. Radially inward facing surface **28** further comprises flange **32** extending radially inward therefrom. In some embodiments, flange **32** is arranged immediately adjacent end **24**. Flange **32** is operatively arranged to engage shoulder **87** to connect tube **80** to connector body **40**. Flange **32** comprises diameter D2. In some embodiments, diameter D1 is greater than diameter D2.

(20) As best seen in FIG. **4**, flange **32** is spaced apart from flange **30** by a first distance. Groove **54** is spaced apart from end **44** by a second distance. Shoulder **87** comprises a width equal to a third distance. The first distance is substantially equal to the sum of the second distance and the third distance. This design results in, when retainer **20** is connected to connector body **40** and tube **80**, flange **32** abutting against surface **88**, as well as surface **86** abutting against end **44** (i.e., retainer **20** clamps shoulder **87** to end **44**). In some embodiments, the first distance is less than the sum of the second distance and the third distance. This results in shoulder **87** and connector body **40** being squeezed or clamped together. In some embodiments, the first distance is greater than the sum of the second distance and the third distance such that shoulder **87** is engaged or in close proximity with connector body **40**. Additionally, flange **30** comprises a first width and groove **54** comprises a second width, the second width being greater than the first width. As such, when retainer **20** is connected to connector body **40** without tube **80** arranged therein, retainer **20** is capable of sliding or displacing in axial direction AD1 and axial direction AD2 with respect to connector body **40**. When tube **80** is properly secured to connector body **40** by retainer **20**, retainer **20** is not capable of displacing in axial direction AD1 and axial direction AD2 with respect to connector body **40**. This feature provides an indicator as to the connection status of fluid connection assembly **10** and will be described in greater detail below. In some embodiments, the first width is substantially equal to the second width.

(21) As previously described, flange **32** comprises diameter D2, which is less than the diameter of shoulder **87**. This design prevents shoulder **87** from being removed from connector body **40**, or tube **80** from displacing in axial direction AD2 or rotating (i.e., bending moment) relative to connector body **40**. Tube **80** may still rotate in a circumferential direction with respect to connector body **40** and retainer **20**.

(22) Section **20A** comprises male connector **36** and section **20B** comprises female connector **38**. As shown, male connector **36** on section **20A** is arranged to engage female connector **38** on section **20B** such that sections **20A** and **20B** are fixedly secured. In some embodiments, male connector **36** is hook-shaped (in a radially outward direction) and includes groove **36A** and one or more radially outward extending projections **36B**. Groove **36A** is arranged in radially outward facing surface **26**. Projections **36B** extend generally radially outward in radial direction RD1 from groove **36A**. In some embodiments, projections **36B** comprise a tapered section near its radially outermost end operatively arranged to allow engagement of projections **36B** with female connector **38**, specifically, apertures **38A**, to occur with greater ease.

(23) Female connector **38** comprises one or more apertures **38A** extending radially inward from radially outward facing surface **26**. Apertures **38A** are operatively arranged to engage projections **36B** to lock section **20B** to section **20A**. Female connector **38** further comprises recess **38B**. In some embodiments, recess **38B** is arranged in radially inward facing surface **28** (i.e., extends radially outward from radially inward facing surface **28**). As section **20B** is displaced toward section **20A**, female connector **38** engages projections **36B** and displaces radially outward in radial direction RD1. Once apertures **38A** are aligned with projections **36B**, female connector **38** snaps back radially inward, in radial direction RD2, thereby securing section **20B** to section **20A**.

(24) Service valve body **100** generally comprises section or tube **102**, section or tube **118**, and

section or tube **130** to form a plurality of ports, for example, port **108**, port **110**, port **124**, and port **136**. Section **102**, section **118**, section **130** are connected such that all ports **108**, **110**, **124**, and **136** are in fluid communication. Section **102** comprises end **104** forming port **108** and end **110** forming port **110**. In some embodiments, and as shown, port **108** and port **110** are concentrically aligned. In some embodiments, port **108** is a service port and may comprise a valve stem or a stop collar (retainer) arranged therein. In some embodiments, section **102** comprises threading **112** on its radially outward facing surface proximate end **104**. In some embodiments, cap **160** is threadably engaged with threading **112** to seal port **108**. In some embodiments, port **110** is a middle port operatively arranged to be connected to a compressor or heat pump in a refrigeration system. In some embodiments, service valve body **100** comprises flange **114** connected to end **106**. Flange **114** may comprise one or more through-holes **116** operatively arranged to secure service valve body **100** to another component (e.g., a compressor). In some embodiments, port **108** comprises a first diameter and port **110** comprises a second diameter, the second diameter being equal to the first diameter. In some embodiments, port **108** comprises a first diameter and port **110** comprises a second diameter, the second diameter being nonequal to the first diameter.

(25) Section **118** comprises end **122** connected to section **102** and end **120** forming port **124**. In some embodiments, port **124** may be a valve port and thus comprise a valve therein (e.g., a Schroeder valve). The valve in port **124** may control fluid flow within service valve body **100**, for example, between ports **108**, **110**, and **136**. In some embodiments, section **118** comprises threading **126** on its radially outward facing surface proximate end **120**. In some embodiments, cap **162** is threadably engaged with threading **126** to seal port **124**. In some embodiments, and as shown, port **124** is concentrically aligned with port **136**. In some embodiments, section **118** is perpendicular to section **102**. In some embodiments, section **118** is non-perpendicular to section **102**.

(26) Section **130** comprises end **134** connected to section **102** and end **132** forming port **136**. In some embodiments, port **136** may be a line port and is operatively arranged to be connected to a component that is filled with fluid or through which fluid flows. For example, port **136** may be connected to a refrigerant line (i.e., tube **80**). In some embodiments, section **130** comprises threading **138** on its radially outward facing surface proximate end **132**. In some embodiments, connector body **40** is threadably engaged with threading **138**, as will be described in greater detail below. In some embodiments, section **130** is perpendicular to section **102**. In some embodiments, section **130** is non-perpendicular to section **102**. In some embodiments, end **132** comprises a frusto-conical or curvilinear taper, wherein the diameter of the radially outward facing surface of section **130** decreases in axial direction AD2 (see FIG. 4). In some embodiments, port **124** comprises a first diameter and port **136** comprises a second diameter, the second diameter being greater than the first diameter. In some embodiments, port **124** comprises a first diameter and port **136** comprises a second diameter, the second diameter being less than or equal to the first diameter.

(27) Connector body **40** comprises through-bore **41** extending from end **42** to end **44**, radially inward facing surface **46**, groove **50**, radially outward facing surface **52**, groove **54**, head **58**, and radially inward facing surface **56**. Connector body **40** is arranged to be removably and sealingly connected to service valve body **100**. Connector body **40** may be screwed onto a service valve body **100**, specifically, by threadably engaging threading **60** on radially inward facing surface **56** with threading **138** of section **130**. Connector body **40** may be screwed onto section **130** via head **58** (e.g., using a wrench). In some embodiments, head **58** is hexagonal; however, it should be appreciated that head **58** may comprise any geometry suitable for applying torque to connector body **40**. It should be appreciated that fluid connection assembly **10** may be used in various components, assemblies, and subassemblies in which fluid connection is desired, for example, refrigeration systems or compressors, or a transmission.

(28) End **42** is connected to section **130**. Connector body **40** is arranged to be connected to a component that is filled with a fluid or through which fluid flows. For example, connector body **40** may be connected to a refrigerant line (i.e., tube **80**). Seals **62A-B** are arranged in connector body

40. Specifically, seal **62A-B** are arranged in grooves **50A-B** to engage tube **80** (i.e., radially outward facing surface **84**). Grooves **50A-B** are arranged spaced apart axially in radially inward facing surface **46**. In some embodiments, seals **62A-B** are O-rings. In some embodiments, connector body **40** only comprises one groove in its radially inward facing surface **46**, and one seal arranged therein. In some embodiments, radially inward facing surface **46** is a substantially cylindrical surface. End **44** is operatively arranged to engage shoulder **87**, specifically, to prevent axial displacement of tube **80** in axial direction **AD1**.

(29) In some embodiments, and as shown, connector body **40** comprises radially inward facing surface **48**, which comprises a diameter that is less than the diameter of radially inward facing surface **46**. Radially inward facing surface **48** is connected radially inward facing surface **146** via surface **47**. In some embodiments, surface **47** is a substantially axial facing surface and faces axial direction **AD2**. In some embodiments, surface **47** acts as a stop or seat for tube **80** thereby preventing tube **80** from being displaced in axial direction **AD1** with respect to connector body **40**. In some embodiments, and as shown, connector body **40** further comprises radially inward facing surface **49** which connects radially inward facing surface **48** with radially inward facing surface **56**. Surface **49** acts as a stop or seat for section **130**. In some embodiments, radially inward facing surface **49** is a frusto-conical surface that increases in diameter in axial direction **AD1** (see FIG. 4). In some embodiments, radially inward facing surface **56** comprises a diameter that is greater than radially inward facing surface **46**. In some embodiments, radially inward facing surface **56** comprises a diameter that is less than or equal to radially inward facing surface **46**.

(30) Groove **54** is arranged in radially outward facing surface **52**. Groove **54** is arranged axially between end **44** and end **42**. Groove **54** comprises a radially outward facing surface and two axial facing surfaces. The radially outward facing surface of groove **54** comprises a diameter that is less than the diameter of radially outward facing surface **52**. In some embodiments, groove **54** is arranged axially between and spaced apart from end **44** and head **58**. In some embodiments, and as shown, groove **54** is arranged immediately adjacent head **58**. Groove **54** is operatively arranged to engage with flange **30** to connect retainer **20** to connector body **40**.

(31) Groove **54** comprises a width that is greater than the width of flange **30**. In the partially assembled locked state, when retainer **20** is connected to connector body **40**, without tube **80** properly connected or present at all, flange **30** and thus retainer **20** is displaceable in axial direction **AD1** and axial direction **AD2** with respect to groove **54** and thus connector body **40** (i.e., flange **30** can slide back and forth within groove **54**). This provides an indication that fluid connection assembly **10** is not properly assembled. When retainer **20** is connected to connector body **40**, with tube **80** properly connected, flange **30** and thus retainer **20** is not displaceable (or is not displaceable as much) in axial direction **AD1** and axial direction **AD2** with respect to groove **54** and thus connector body **40** (i.e., flange **30** abuts against the left surface of groove **54** as shown in FIG. 4). In some embodiments, connector body **40** comprises a metal, a polymer, and/or a ceramic.

(32) To assemble fluid connection assembly **10**, connector body **40** is connected to service valve body **100**, specifically, section **130**. For example, connector body **40** may be screwed onto section **130** such that threading **60** is engaged with threading **138**. In some embodiments, end **132** (or its frusto-conical taper) is engaged with radially inward facing surface **49**. In some embodiments, end **132** is spaced apart from radially inward facing surface **49**. Tube **80** is inserted in axial direction **AD1**, with end **82** first, into connector body **40**. Radially outward facing surface **84** engages seals **62A-B** and section **83** is arranged inside of connector body **40** proximate radially inward facing surface **46**. Shoulder **87** engages end **44** of connector body **40**, specifically, surface **86** abuts against end **44**. In some embodiments, and as shown, shoulder **87** resides completely outside of connector body **40**. In some embodiments, end **82** abuts against surface **47**. In some embodiments, end **82** is spaced apart from surface **47**. Then, retainer **20** is secured over both connector body **40** and tube **80**. Specifically, sections **20A** and **20B** are arranged over connector body **40** such that flange **30** is at least partially aligned with groove **54** and flange **32** is at least partially aligned with surface **88**.

Sections **20A** and **20B** are displaced radially inward toward each other (i.e., in radial direction **RD2**) until female connector **38** fully engages male connector **36** and retainer **20** is in the locked state. Additionally, flange **32** engages tube **80**, specifically radially outward facing surface **90**, thereby preventing the ingress of foreign materials into fluid connection assembly **10**.

(33) In the fully assembled locked state, flange **30** engages groove **54**, radially inward facing surface **28** engages radially outward facing surface **52** and flange **32** engages shoulder **87** of tube **80**. In some embodiments, in the fully assembled locked state, flange **30** abuts against the left surface of groove **54**, flange **32** abuts against surface **88**, and surface **86** abuts against end **44**. In some embodiments, in the fully assembled locked state, flange **30** abuts against the radially outward facing surface of groove **54**, thus preventing radial or circumferential displacement of retainer **20** with respect to connector body **40**. In some embodiments, flange **32** engages radially outward facing surface **90**. The engagement of tube **80** with connector body **40** prevents axial displacement of tube **80** in axial direction **AD1** and the engagement of retainer **20** with connector body **40** and tube **80** prevents axial displacement of tube **80** in axial direction **AD2**, as well as radial directions **RD1** and **RD2**, relative to connector body **40**.

(34) To disassemble, female connector **38** is displaced radially outward in radial direction **RD1** with respect to male connector **36** until apertures **38A** disengages projections **36B**. Sections **20A** and **20B** are then separated to disengage flange **32** from shoulder **87**, at which point tube **80** can be removed from connector body **40**.

(35) It will be appreciated that various aspects of the disclosure above and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

LIST OF REFERENCE NUMERALS

(36) **10** Fluid connection assembly **20** Retainer **20A** Section **20B** Section **21** Hole **22** End **24** End **26** Radially outward facing surface **28** Radially inward facing surface **30** Flange **32** Flange **34** Hinge **36** Male connector **36A** Groove **36B** Projection(s) **38** Female connector **38A** Aperture(s) **38B** Recess **40** Connector body **41** Through-bore **42** End **44** End **46** Radially inward facing surface **47** Axial surface or seat **48** Radially inward facing surface **49** Radially inward facing surface **50A** Groove **50B** Groove **52** Radially outward facing surface **54** Groove **56** Radially inward facing surface **58** Head **60** Threading **62A** Seal **62B** Seal **80** Tube **82** End **83** Section **84** Radially outward facing surface **86** Surface **87** Shoulder or bead **88** Surface **89** Section **90** Radially outward facing surface **92** End **94** Through-bore **100** Service valve body **102** Section or tube **104** End **106** End **108** Port or service port **110** Port or middle port **112** Threading **114** Flange **116** Hole(s) **118** Section or tube **120** End **122** End **124** Port or valve port **126** Threading **130** Section or tube **132** End **134** End **136** Port or line port **138** Threading **160** Cap **162** Cap **AD1** Axial direction **AD2** Axial direction **RD1** Radial direction **RD2** Radial direction **D1** Diameter **D2** Diameter

Claims

1. A fluid connection assembly, comprising: a connector body, comprising: a first end; a second end; a through-bore; a first radially outward facing surface comprising an annular groove; a first radially inward facing surface extending from the first end; and a second radially inward facing surface extending from the second end; and a retainer operatively arranged to be removably connected to the connector body, the retainer including: a third end; a fourth end; a first section comprising a male connector component; a second section displaceable with respect to the first section, the second section comprising a female connector component operatively arranged to engage the male connector component to fixedly secure the second section to the first section; a third radially inward facing surface; a first flange extending radially inward from the third radially

inward facing surface and operatively arranged to engage the annular groove, the first flange comprising a first innermost diameter; and a second flange extending radially inward from the third radially inward facing surface, the second flange comprising a second innermost diameter, less than the first innermost diameter.

2. The fluid connection assembly as recited in claim 1, wherein the first radially inward facing surface comprises at least one annular groove.

3. The fluid connection assembly as recited in claim 2, wherein the at least one annular groove comprises: a first annular groove; and a second annular groove spaced apart in an axial direction from the first annular groove.

4. The fluid connection assembly as recited in claim 2, wherein a seal is arranged in the at least one annular groove.

5. The fluid connection assembly as recited in claim 1, wherein the second radially inward facing surface comprises threading.

6. The fluid connection assembly as recited in claim 1, wherein the connector body further comprises a fourth radially inward facing surface arranged between the first radially inward facing surface and the second radially inward facing surface, wherein a diameter of the fourth radially inward facing surface is less than both the first radially inward facing surface and the second radially inward facing surface.

7. The fluid connection assembly as recited in claim 6, wherein the connector body further comprises an axial surface connecting the first radially inward facing surface and the fourth radially inward facing surface.

8. The fluid connection assembly as recited in claim 1, wherein: the first flange is arranged immediately proximate the third end; and the second flange is arranged immediately proximate the fourth end.

9. The fluid connection assembly as recited in claim 1, further comprising a tube including a shoulder, wherein the retainer is operatively arranged to secure the tube to the connector body.

10. The fluid connection assembly as recited in claim 9, wherein in a connected state, the shoulder and the connector body are squeezed together by the retainer.

11. The fluid connection assembly as recited in claim 1, further comprising a service valve body comprising a plurality of ports, wherein the connector body is operatively arranged to be removably connected to at least one port of the plurality of ports.

12. A fluid connection assembly, comprising: a connector body, comprising: a first end; a second end; a through-bore; a first radially outward facing surface comprising an annular groove; a first radially inward facing surface extending from the first end; and a second radially inward facing surface extending from the second end; and a retainer operatively arranged to be removably connected to the connector body, the retainer including: a third end; a fourth end; a second radially outward facing surface; a third radially inward facing surface; a first section comprising a male connector component fixedly secured to the second radially outward facing surface; a second section displaceable with respect to the first section, the second section comprising a female connector component including a through-hole operatively arranged to engage the male connector component to fixedly secure the second section to the first section; a first flange extending radially inward from the third radially inward facing surface and operatively arranged to engage the annular groove; and a second flange extending radially inward from the third radially inward facing surface.

13. The fluid connection assembly as recited in claim 12, wherein: the male connector component extends radially outward from the second radially outward facing surface; and the second section is hingedly connected to the first section.

14. The fluid connection assembly as recited in claim 12, wherein: the first flange is arranged immediately proximate the third end; and the second flange is arranged immediately proximate the fourth end.

15. The fluid connection assembly as recited in claim 12, wherein: the first flange comprises a first diameter; the second flange comprises a second diameter; and the first diameter is greater than the second diameter.
16. The fluid connection assembly as recited in claim 12, wherein the first radially inward facing surface comprises: a first annular groove; and a second annular groove spaced apart in an axial direction from the first annular groove.
17. The fluid connection assembly as recited in claim 12, wherein the second radially inward facing surface comprises threading.
18. The fluid connection assembly as recited in claim 12, wherein the connector body further comprises: a third radially inward facing surface arranged between the first radially inward facing surface and the second radially inward facing surface, the third radially inward facing surface having a diameter that is less than a diameter of both the first radially inward facing surface and the second radially inward facing surface; and an axial surface connecting the first radially inward facing surface and the third radially inward facing surface.
19. The fluid connection assembly as recited in claim 18, wherein the connector body comprises a head arranged at the second end.
20. A fluid connection assembly, comprising: a connector body, comprising: a first end; a second end; a through-bore; a first radially outward facing surface comprising an annular groove; a first radially inward facing surface extending from the first end; and a second radially inward facing surface extending from the second end; a retainer operatively arranged to be removably connected to the connector body, the retainer including: a third end; a fourth end; a first section comprising a male connector component; a second section displaceable with respect to the first section, the second section comprising a female connector component operatively arranged to engage the male connector component to fixedly secure the second section to the first section; a third radially inward facing surface; a first flange extending radially inward from the third radially inward facing surface and operatively arranged to engage the annular groove; and a second flange extending radially inward from the third radially inward facing surface; and a tube including a shoulder, wherein the retainer is operatively arranged to secure the tube to the connector body; wherein in a connected state, the shoulder and the connector body are squeezed together by the retainer.
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